

Shanrong Wu

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, Illinois

Bachelor of Science in Computer Science — GPA: 3.88

Expected Graduation Date: May 2028

Relevant Coursework: Introduction to Computer Science I, Introduction to Computer Science II (Honors), Discrete Structures, Data Structures, Computer Architecture, Software Design Lab

PROJECTS

HackIllinois 2026 – HackAstra | *React, Python, Flask, Gemini API, Supermemory*

Feb 2026

- Led a team of 3 to develop an AI inspection software application for Caterpillar Track, automating daily inspections through audio, video, and image-based input.
- Integrated the Gemini API to deliver real-time AI generated inspection feedback during the inspection workflow.
- Built full stack features using React and Flask, including inspection processing and reporting.
- Implemented a machine inspection history to store and display previous inspection results for machines.
- Improved report generation by debugging inspection output and creating functionality to consolidate multiple same day inspections into one unified report in daily checklist and history.

Raytracer in Rust | *Rust*

Oct 2025 – Jan 2026

- Developed a ray tracing application in Rust with a 3 person team, utilizing Rust crates for rendering and systems level performance improvements.
- Designed and implemented geometry processing utilities to convert 3D coordinates and STL mesh data into triangle objects for ray object intersection testing using the Möller–Trumbore algorithm.
- Optimized runtime by about 10x with parallel processing, multithreading, and denoising techniques.
- Created scenes with STL based models and geometric objects to check rendering accuracy and scene composition.

IlliniBites | *TypeScript, React Native, Expressjs, HTML/CSS*

Sep 2025 – Dec 2025

- Developed a mobile app called IlliniBites with a team using React Native framework.
- Improved user experience by implementing filtering and sorting feature on specific topics.
- Designed and implemented a backend API with Express.js for efficient retrieval and delivery of restaurant data.

Paradox | *JavaScript, React, Next.js, HTML/CSS, TailwindCSS, Supabase*

Jul 2025 – Aug 2025

- Developed an interactive web application using Next.js, React, and JavaScript to track and manage multiple data attributes for the Paradox game clan.
- Improved data logging efficiency by moving from manual Google Sheets workflows to a Supabase database.
- Built data visualizations to display user and clan metrics, and designed a responsive user interface using HTML/CSS and Tailwind CSS.

EXTRACURRICULARS

SIGmobile

Oct 2025 – Present

Backend Developer

Urbana, Illinois

- Contribute to backend for CS Course Recommender, a mobile app designed to recommend CS courses to students.
- Collected and processed course data by scraping UIUC Course API and assigning interest tags to each course.

SIGrobotics - F1tenth

Sep 2025 – Present

Simulation/Planning Group

Urbana, Illinois

- Develop autonomous driving functionality for the F1TENTH racecar in the F1TENTH Gym simulation using C++, ROS 2, PID control, and path planning/control algorithms across multiple track layouts.
- Support development for the physical racecar by working with 2D LiDAR, stereo vision, and Jetson Nano.
- Participate in meetings to collaborate on project progress and continued learning of autonomous racing concepts.
- Help define project goals and contribute to software development for both simulated and physical racecar testing.
- Engineered a local planner in Python and ROS2 Humble that integrated global planner output with robot pose data to generate waypoints for pure pursuit based navigation.

TECHNICAL SKILLS

Languages: C++, Java, JavaScript, TypeScript, HTML/CSS, LabVIEW, Rust, Python

Frameworks: React, Next.js, TailwindCSS, ROS2

Developer Tools: Linux, Git, GitHub, GitLab, Docker, VS Code, IntelliJ, Supabase